Amadeo Huerta Moncho

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AI/ML researcher and engineer with a background in computer vision, probabilistic machine learning, and scalable AI systems. Experienced in developing reproducible ML pipelines across research and applied settings, spanning deep learning, representation learning, and intelligent systems. Interested in robust and efficient machine learning, from theoretical foundations to real-world applications.

Education

MSc IN INTELLIGENT INTERACTIVE SYSTEMS | 2026 | UNIVERSITAT POMPEU FABRA

- Focus: Probabilistic Machine Learning, Bayesian Inference, Self-supervised Learning, Computer Vision and Uncertainty Quantification.
- Master's thesis: On Probabilistic Prediction and Information-Theoretic Operators (9.8/10).

BSc IN ARTIFICIAL INTELLIGENCE | 2021-2025 | UNIVERSITAT POLITÈCNICA DE CATALUNYA

- Focus: Deep Learning, Computer Vision and Data-driven Systems.
- Related coursework: Robotics, Human Language Processing, Reinforcement Learning, Data Science, Optimization and Statistical Modelling.
- Erasmus+ International Mobility: Odense, University of Southern Denmark (SDU), 2024

Experience

RESEARCH ASSISTANT | AI & ML GROUP, UNIVERSITAT POMPEU FABRA | MAR 2026 – SEP 2026

- Conducted research on predictive information gain and information-theoretic methods for efficient self-supervised learning.
- Developed Bayesian and neural masked-prediction models, implementing reproducible experiments and evaluation pipelines in Python/PyTorch.
- Evaluated predictive uncertainty and training efficiency through controlled experiments, comparing information-based selection strategies against reference baselines.

MEDICAL VISION RESEARCHER (INTERNSHIP) | IOC-UPC | JAN 2025 – JUL 2025

- Developed computer vision pipelines for radiographic analysis, including segmentation characterization and quantitative assessment.
- Combined deep learning and classical vision methods for robust medical image processing.

Presentations

- GenU 2026 Workshop, Copenhagen — Poster presentation on predictive information gain and masked self-supervised learning.

Skills

APPLIED AI/ML RESEARCH

- **Research Areas:** Probabilistic Machine Learning, Self-Supervised Learning, Computer Vision, Representation Learning and Uncertainty Quantification.
- **Methods:** Bayesian Inference, Information Theory, Deep Learning, Statistical Analysis and Experimental Design.
- **ML Systems**: GPU Computing, Parallel Computing, Performance Benchmarking and Reproducible ML Experimentation.

TECHNICAL STACK

- **Languages**: Python, C/C++, MATLAB, SQL, R | **Frameworks**: PyTorch, TensorFlow, OpenCV | **Systems**: Git, Spark, Hadoop, ROS/ROS2.
- **Additional AI Areas**: Foundations of Large Language Models (LLMs), Reinforcement Learning, Generative AI.

Selected Projects

- On Probabilistic Prediction and Information-Theoretic Operators — Master's Thesis

Investigated information-theoretic methods using linear and probabilistic models; achieved reference performance with ~33.5% fewer training images in controlled PPCA experiments on MNIST.

- ParaRNN — GPU Performance Benchmarking

Evaluated temporal parallelism and CUDA kernel fusion for recurrent models, measuring up to 221× forward-pass speedup in controlled GPU microbenchmarks.

Additional Information

- Languages: Spanish (Native), Catalan (Native), English (Advanced), French (Basic)
- Certification: Research Ethics and Personal Data Protection — CIREP (UPF, 2025)